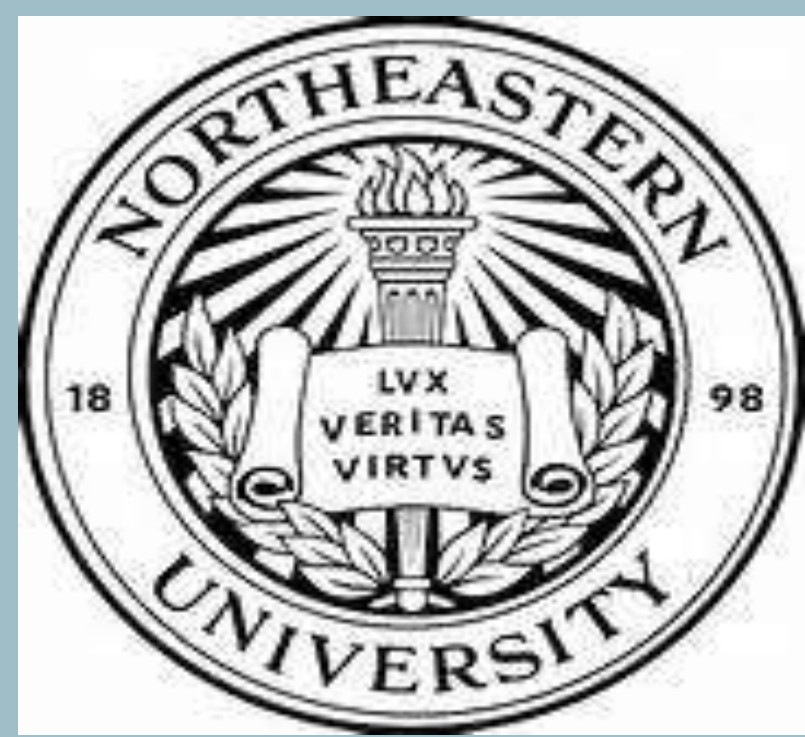




GFP Expression and Purification Process: From Transformation to Chracterization

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Introduction and Objectives

This poster describes the transformation of E. coli with the GFP gene, induction of GFP expression, confirmation via SDS-PAGE and Western blot, purification of GFP using two-step chromatography (drip column and HPLC), identification of purified fractions spectrophotometrically, and characterization of purified GFP's fluorescence under varying temperature and pH conditions, presenting the full workflow, data, and observations.

Background/ Literature Review

This study by Dong et al. (2019) presents an economical and eco-friendly approach for producing and purifying recombinant green fluorescent protein (GFP) using a plant-based expression system. *Nicotiana benthamiana* leaves were utilized for high-level transient overexpression of GFP using a Tobacco mosaic virus RNA-based vector. The novel purification strategy combined an alcohol/salt aqueous two-phase system (ATPS) with hydrophobic interaction chromatography (HIC). In contrast to this study, we aim on efficient purification of GFP in E.coli, differing in purification strategy, purpose and engineering mechanisms.

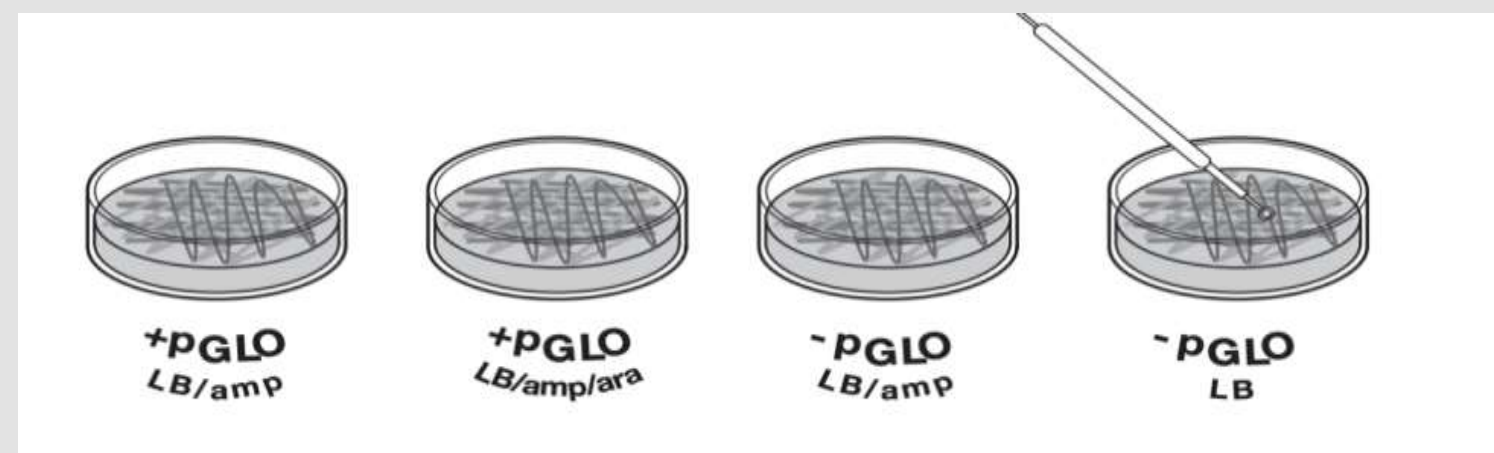
Methods

1. E. coli cells were made competent and transformed with the pGLO plasmid containing the GFP gene via heat shock. Transformed colonies were selected on ampicillin plates, isolated on streak plates, and cultured in liquid LB media. Induced GFP expression was confirmed by visualizing green fluorescence under UV light.

(DAY -1)	DAY 0	DAY 1	DAY 2	DAY 3
	Your Class Session - Transform E. coli and Spread Plates	Incubate 37C ~48 hrs	Determine Transformation Efficiency Shake 37C ~24 hrs Innoculate Liquid Culture	Cryopreserve

3. Protein was extracted, centrifugated, and hydrophobic interaction chromatography (HIC) was carried out to purify GFP protein from bacterial cells. We evaluated its fluorescent properties under UV light and stability under temperature denaturation.

2. Transformed E. coli cells containing the pGLO plasmid were selected on ampicillin plates. Isolated colonies were then inoculated into liquid LB media and grown overnight with shaking at 37°C to generate enriched cultures.

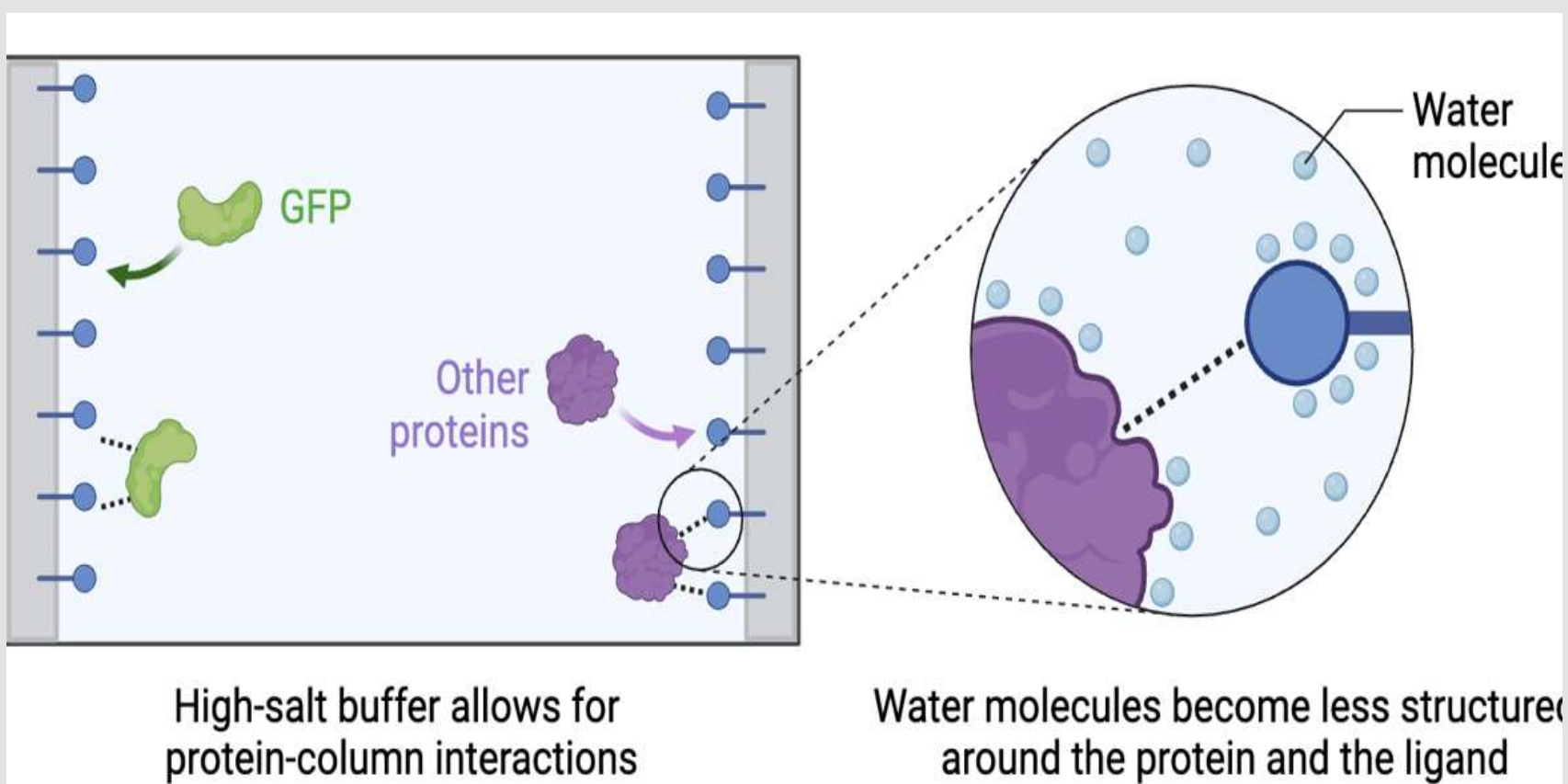


4. GFP was tested for stability under pH changes and temperature denaturation, and the effect of pH changes on GFP fluorescence and the temperature at which GFP denatures were evaluated by observing its fluorescence under UV light.

Methods

5. The AKTA Pure HPLC system was used for high-resolution separation and purification of GFP. Samples were injected into the system and separated on columns using precise control of mobile phase gradients. Eluted compounds were detected by monitoring UV absorbance and collected in purified fractions.

6. pGLO proteins were identified using antibodies, involving gel electrophoresis, western blotting and Ponceau staining.



Results

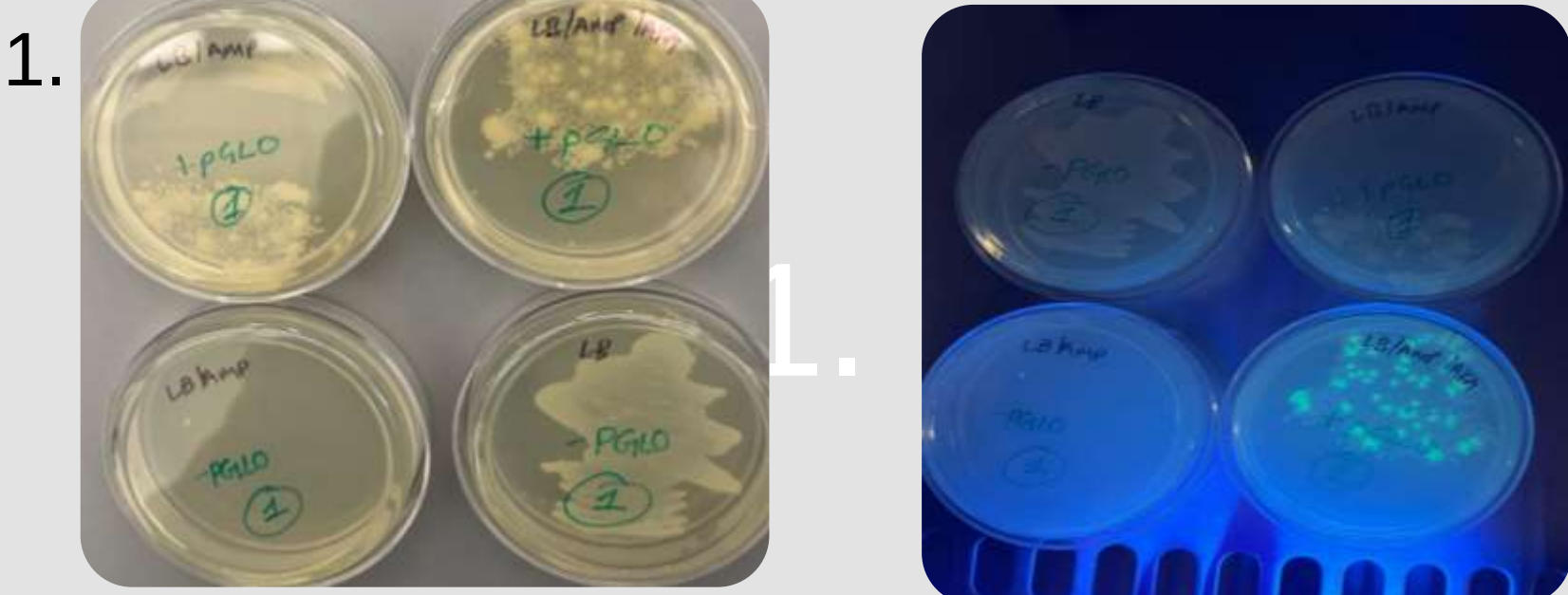


Figure 1 (a) E.Coli under normal room light
Figure 1 (b) E.Coli under U.V Lights

Results

2. Transformation was successful, as evidenced by colony growth on +pGLO AMP plates and no growth on -pGLO AMP plates. Additionally, the fluorescent colonies on the +pGLO/AMP/ARA plate indicate successful insertion and expression of the GFP gene under the Arabinose promoter.

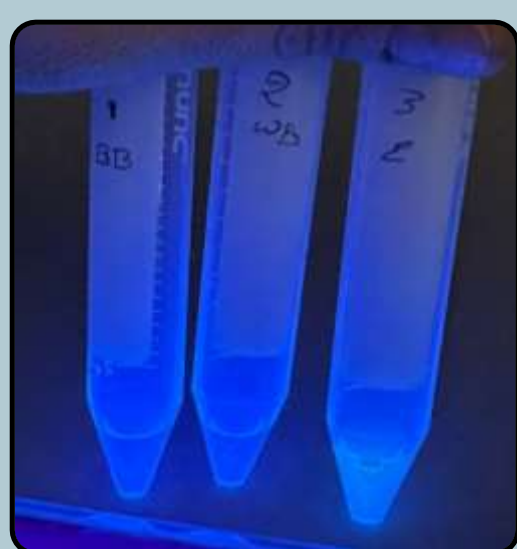


Figure 2 (a) Purified GFP was eluted via HIC, which is seen in the eluted column (first from right)



Figure 2 (b) GFP Stability Study under UV Lights at pH 4, 7 and 10 (L to R)



Figure 2 (c) GFP Stability Study under UV Lights at Temperatures - 50, 75 and 100 (L to R)

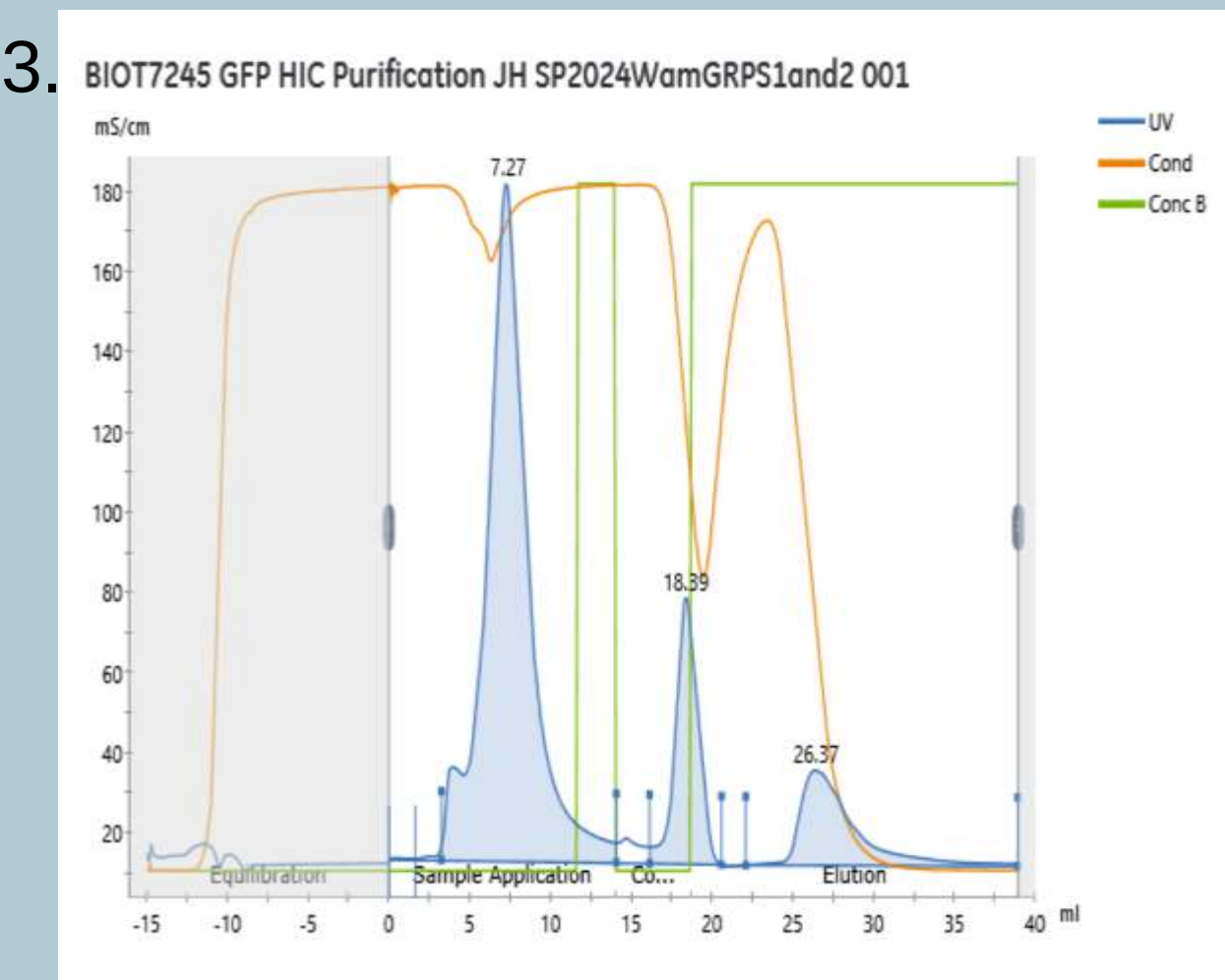
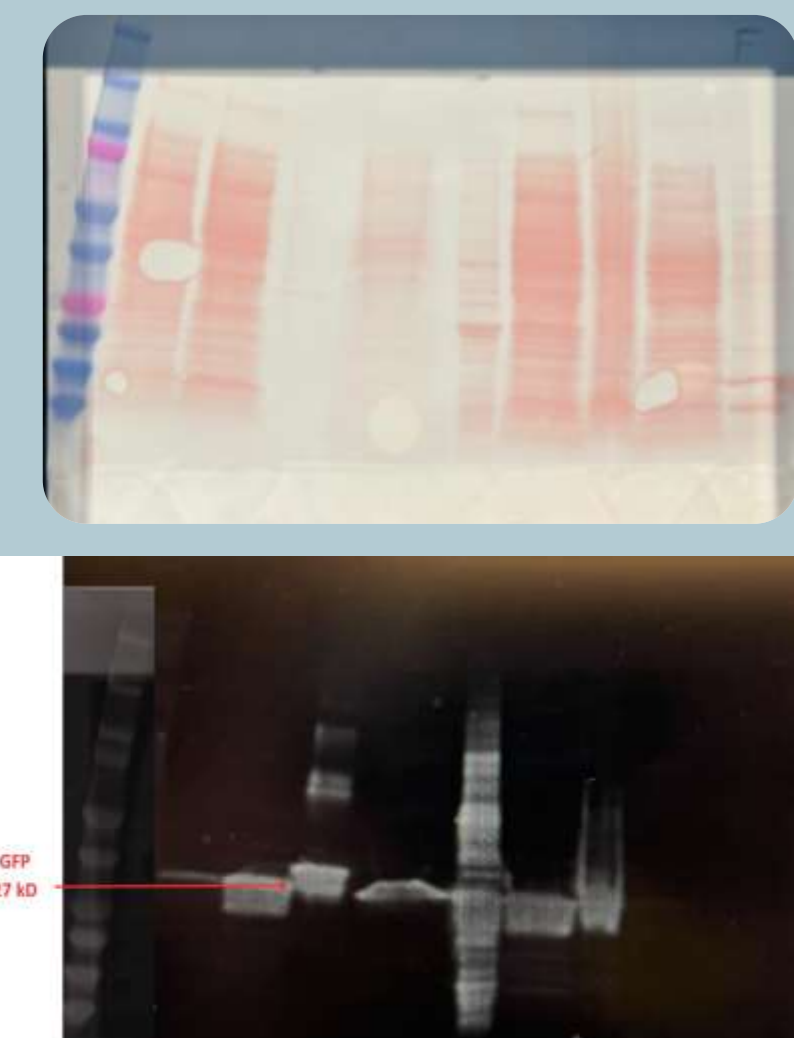


Figure 3 GFP purification carried out via AKTA pure. The percentage of protein eluted as seen in elution peaks B and C, is 28.29%. The dip in conductivity shift depicts the change from high salt to low salt buffer concentration.

Figure 4 (a) Blot Membrane stained by Ponceau Dye.

Figure 4 (b) GFP visualized under Western Blot.

As anticipated, the +pGLO, positive control, and HIC elute has GFP and aligns with each other at 27 kD (The actual weight of GFP) when compared with the ladder.



Conclusion

The E. coli transformation with the GFP gene was successful, evident from colony growth and fluorescence. GFP was effectively purified using chromatography techniques. Its stability under varying pH and temperature conditions was characterized. Western blot confirmed purified GFP identification. Overall, the experiment achieved its objectives of GFP expression, purification, and characterization. Future work could explore GFP applications and engineering for enhanced properties.

Acknowledgements

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References

Dong, J., Ding, X., & Wang, S. (2019). Purification of the recombinant green fluorescent protein from tobacco plants using alcohol/salt aqueous two-phase system and hydrophobic interaction chromatography. BMC Biotechnology, 19(1). <https://doi.org/10.1186/s12896-019-0590-y>

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